WORKSHEET 9.6

Free-Response Type: Particle Motion

Unit 9: Exam Strategy and Free-Response Types

Key ideas

- The particle moves right when $v > 0$ and left when $v < 0$. It changes direction where v changes sign.
- Speed is increasing when v and a have the same sign.
- Position: $x(b) = x(a) + \int_a^b v(t)\,dt$. Total distance: $\int_a^b |v(t)|\,dt$.
- For "farthest left" or "farthest right", compare positions at the endpoints and at each change of direction.

Worked examples

Example 1 · Changes of direction

A particle has $v(t) = t^2 - 5t + 4$ for $0 \le t \le 6$ and $x(0) = 2$. When does it change direction?

Solution. $v(t) = (t-1)(t-4)$ changes sign at both zeros.

At $t = 1$ and $t = 4$.

Example 2 · Speeding up or slowing down

Is the particle speeding up or slowing down at $t = 2$?

Solution. $v(2) = -2$ and $a(2) = 2(2) - 5 = -1$.

Speeding up: v and a have the same sign.

Example 3 · Position

Find the position of the particle at $t = 4$.

Solution. $2 + \int_0^4 (t^2 - 5t + 4)\,dt = 2 + \left(\frac{64}{3} - 40 + 16\right)$.

$x(4) = -\frac{2}{3}$.

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Practice

CORE

1. A particle has $v(t) = 12 - 3t^2$ for $t \geq 0$. When is it at rest?

2. Find the acceleration of that particle at $t = 1$.

3. Find the displacement of that particle for $0 \leq t \leq 2$.

4. Find the total distance that particle travels for $0 \leq t \leq 3$.

EXAM LEVEL

5. *Calculator.* A particle has $v(t) = \ln(t^2 + 1) - 1.5$ for $0 \leq t \leq 5$, and $x(0) = 4$. Find the time at which it changes direction. Justify your answer.

6. *Calculator.* Find the acceleration of that particle at $t = 3$. Is its speed increasing or decreasing then?

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7. *Calculator.* Find the position of that particle at $t = 5$.

8. *Calculator.* Find the total distance that particle travels for $0 \leq t \leq 5$.

Question 9 · Free-response practice

No calculator. A particle moves along the x-axis with velocity $v(t) = 6 - 3\sqrt{t}$ for $0 \leq t \leq 9$. At time $t = 0$ its position is $x = -5$.

(a) Find the time at which the particle is at rest, and the direction in which it moves before that time.

(b) Find the position of the particle at $t = 9$.

(c) Find the total distance travelled for $0 \leq t \leq 9$, and the acceleration of the particle at $t = 4$.

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Worked solutions

1. $t = 2$.

2. $a(t) = -6t$, so $a(1) = -6$.

3. $\left[12t - t^3\right]_0^2 = 16$.

4. Displacement on $[2, 3]$ is $(36 - 27) - 16 = -7$. Distance $16 + 7 = 23$.

5. $v(t) = 0$ at $t \approx 1.866$, where v changes from negative to positive.

6. $a(3) = \dfrac{2(3)}{10} = 0.6$, and $v(3) \approx 0.803 > 0$. The speed is increasing, since v and a are both positive.

7. $4 + \displaystyle\int_0^5 v(t)\,dt \approx 5.537$.

8. $\displaystyle\int_0^5 |v(t)|\,dt \approx 4.686$.

9. (a) $6 - 3\sqrt{t} = 0$ at $t = 4$. For $0 \le t < 4$, $v > 0$: the particle moves to the right.

   (b) $-5 + \left[6t - 2t^{3/2}\right]_0^9 = -5 + (54 - 54) = -5$.

   (c) Displacement on $[0, 4]$ is $24 - 16 = 8$ and on $[4, 9]$ is -8: distance 16. $a(t) = -\dfrac{3}{2\sqrt{t}}$, so $a(4) = -\dfrac{3}{4}$.

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